

Cognitive Functions and Orientation

From awareness of self and surroundings to body awareness and movement control

COGNITIVE FUNCTIONS

Mental processes used for **thinking, understanding, learning, remembering, attending**, and interpreting information.

🎯 Attention

📖 Memory

👁 Perception

🗣 Language

💡 Reasoning

📋 Executive functions

📍 Orientation

CENTRAL QUESTION



How does a person know who they are, where they are, when it is, and what is happening?

Orientation is a cognitive / mental-status function

Sources: APA, DSM-5-TR (2022; APA updates 2025); Campbell & Barohn, DeJong's The Neurologic Examination, 8e (2020).



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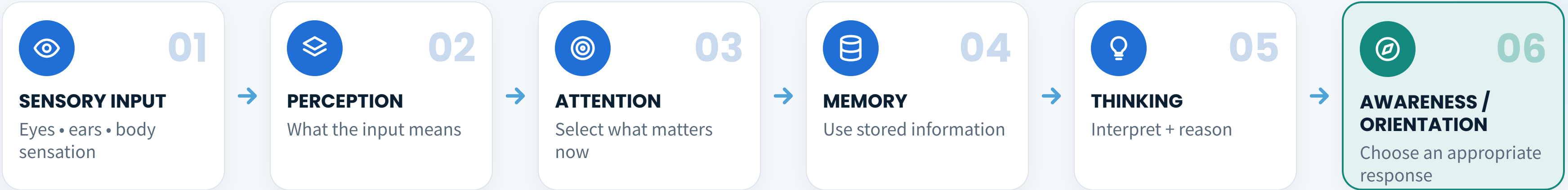


ORIENTATION

Mental-state awareness of person, place, time and situation

How Does a Person Know Where They Are and What Is Happening?

Awareness emerges from integrated information processing — not a single brain function.



CLINICAL EXAMPLE — NEW PATIENT ON THE WARD

Ask:



Who are you?

Person



Where are you?

Place



What day is it?

Time



Why are you here?

Situation

PSYCHIATRY + NEUROLOGY



Psychiatry

Disorientation can occur in delirium and some severe cognitive disorders. It is not a feature of every psychiatric condition.



Neurology

Disorders affecting attention, memory, sensory processing or body-position pathways can alter awareness or performance.



Orientation is the brain's integrated awareness of self + surroundings — not simply “knowing facts”.

Sources: APA, DSM-5-TR (2022; 2025 update); Campbell & Barohn, 8e (2020).

How Do We Assess Orientation?

Bedside mental-status assessment uses conversation, observation and clinical context.

01
PERSON

- ✓ Name?
- ✓ Who are you?
- ✓ Who are the people around you?

02
PLACE

- ✓ Where are you?
- ✓ Which hospital / clinic / home?
- ✓ Which city?

03
TIME

- ✓ Day? Month? Year?
- ✓ Time or part of day?

04
SITUATION

- ✓ Why are you here?
- ✓ What is happening right now?

Interpret responses in context: consider age, education, language, culture, sensory limitations and the current clinical condition.

 Age

 Education

 Language

 Culture

 Sensory limitations

 Current clinical condition

 **Orientation questions are part of the mental status examination; they should not be interpreted in isolation.**

Sources: APA, DSM-5-TR (2022; 2025 update); Campbell & Barohn, 8e (2020).

Why Can Orientation Become Impaired?

Many acute, chronic, neurologic, psychiatric and medication-related factors can contribute.

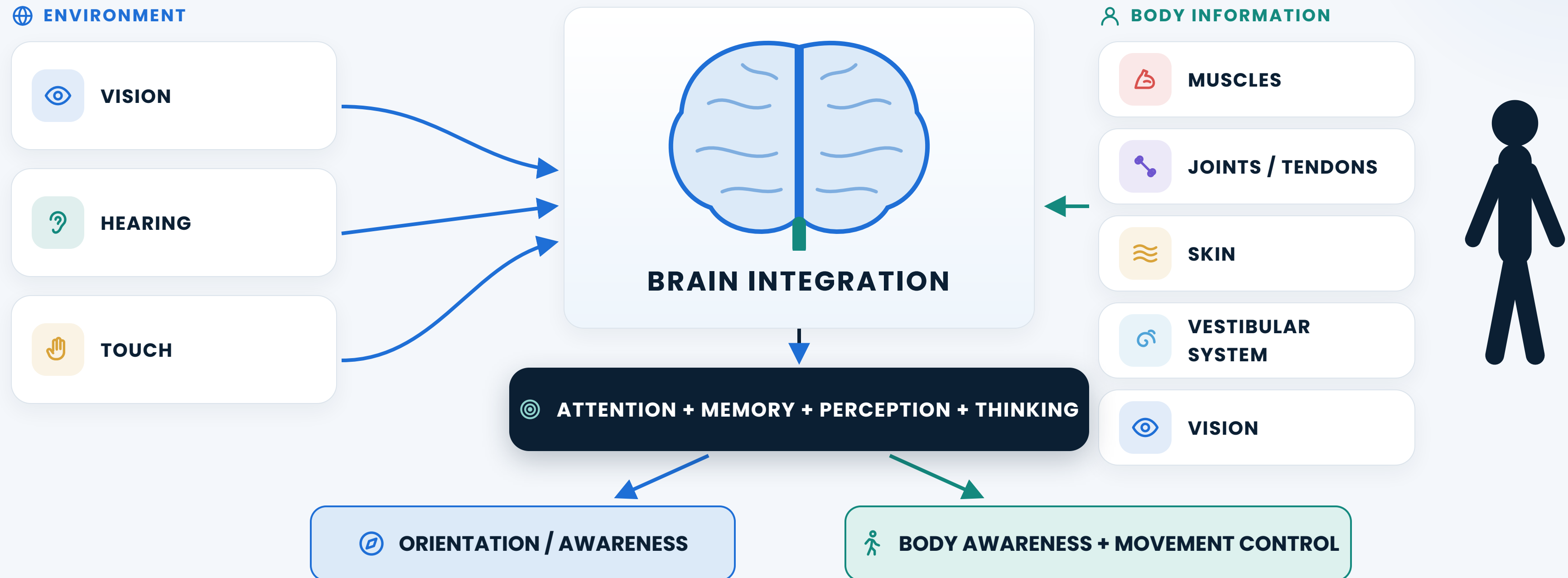


⚠ KEY CLINICAL POINT • Sudden new confusion or disorientation needs assessment for an underlying medical or neurologic cause.

Sources: APA, DSM-5-TR (2022; 2025 update); Campbell & Barohn, 8e (2020).

How the Brain Builds Awareness of Self and Environment

A visual model: environmental information and body information are combined by the nervous system.

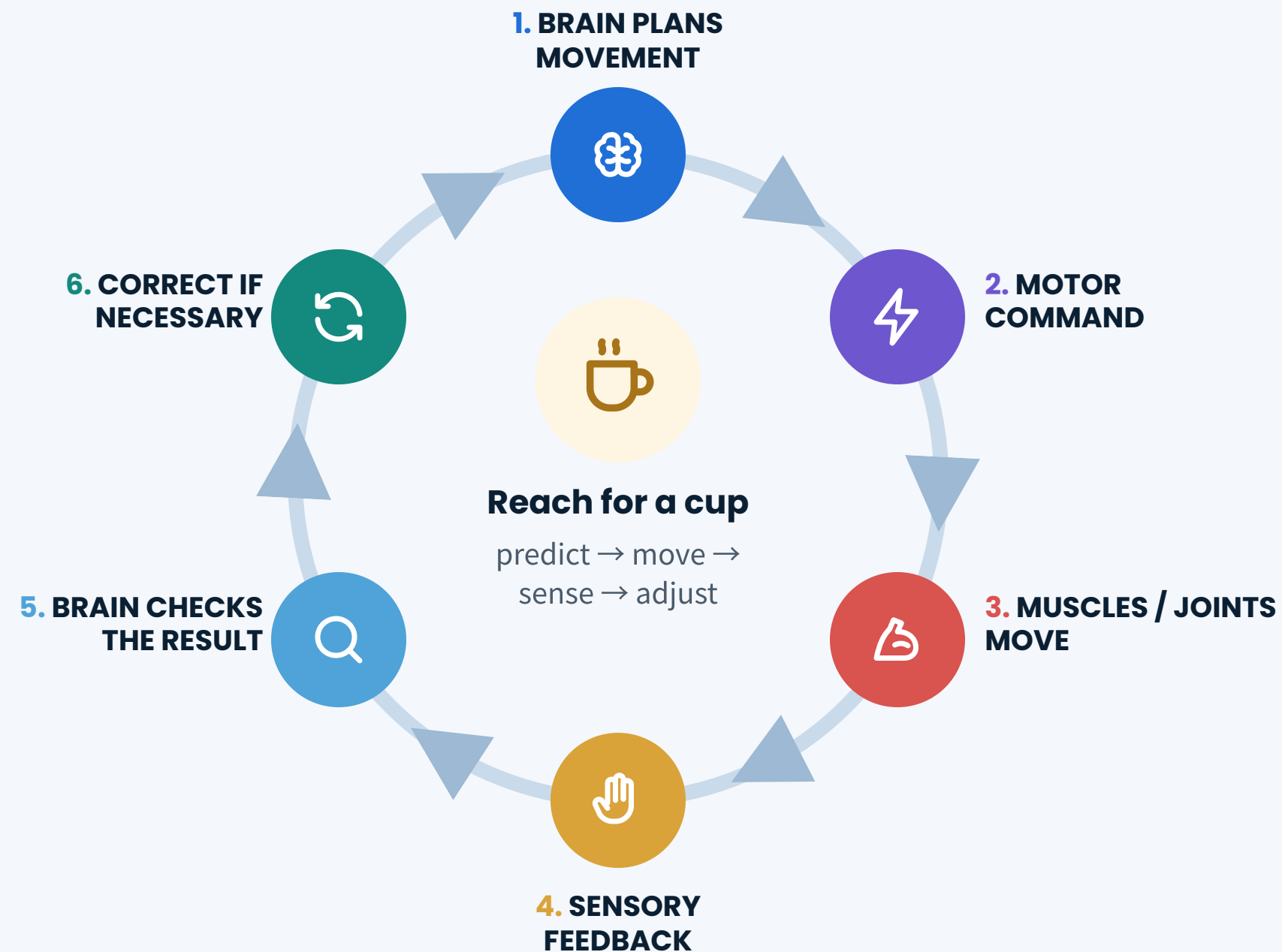


The brain continuously combines information from the environment and the body to guide behaviour.

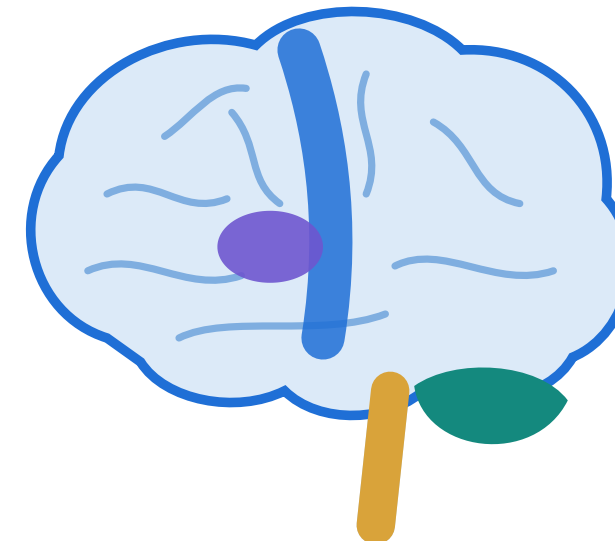
Sources: Campbell & Barohn, 8e (2020); Gilman, J Neurol Neurosurg Psychiatry (2002), joint position sense review.

What Is the Role of Motor Control?

Motor control plans, initiates, coordinates and adjusts movement using sensory feedback.



MAJOR CONTRIBUTORS



- Motor cortex**
Voluntary movement planning + execution
- Cerebellum**
Coordination • timing • error correction • balance
- Basal ganglia**
Movement initiation + regulation
- Spinal cord + peripheral nerves**
Carry motor and sensory signals

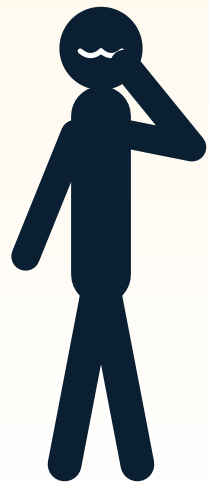
 **Motor control is more than muscle strength; it depends on sensory information and coordination.**

Sources: Campbell & Barohn, DeJong's The Neurologic Examination, 8e (2020).

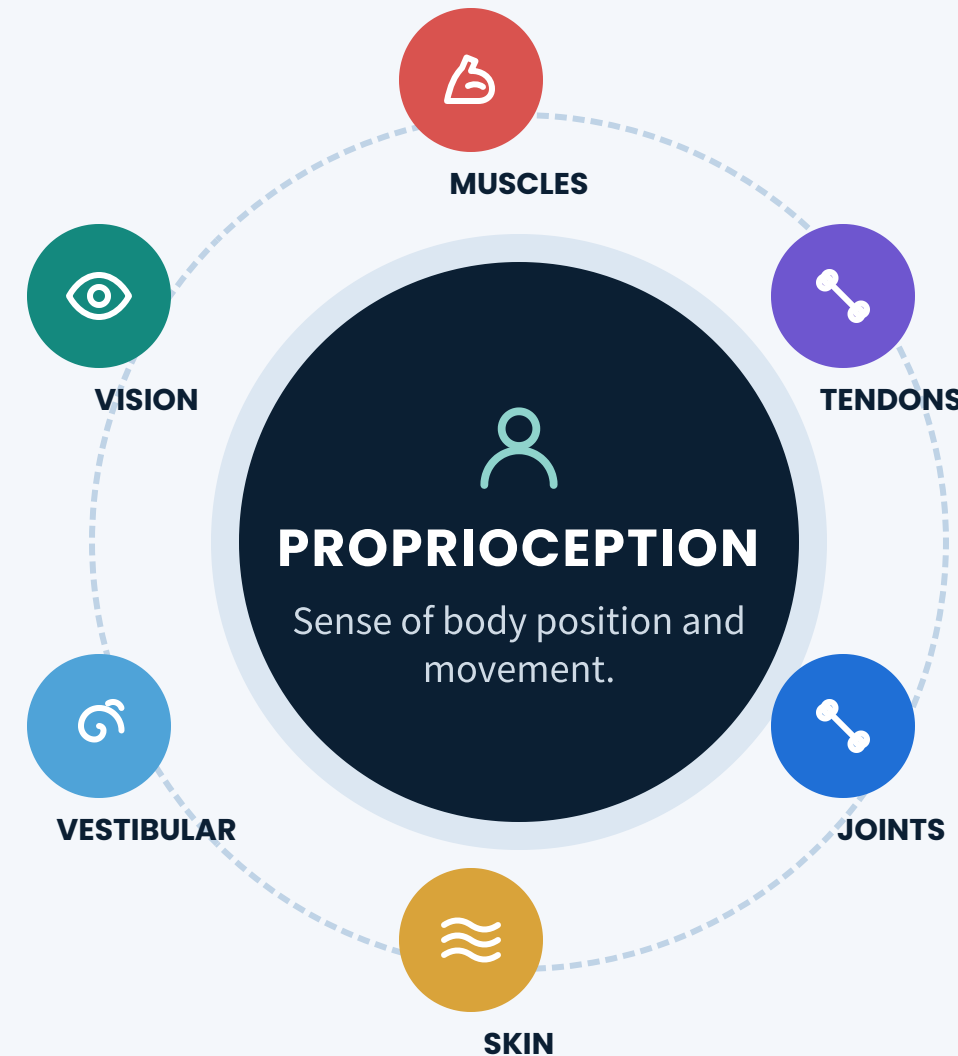
Body Awareness and Proprioception

Question: How can you know where a limb is even when you cannot see it?

CLASSROOM DEMONSTRATION



- 1 Eyes closed
- 2 Raise an arm
- 3 Touch your nose
- 4 Identify limb position without looking



KEEP THE TERMS DISTINCT

Proprioception

Awareness of body position / movement

Kinesthesia

Awareness specifically of movement

Body schema / awareness

Internal representation and awareness of the body

Proprioception → supplies body-position information

Motor control → uses sensory feedback to guide movement

Orientation → concerns person, place, time and situation

Sources: Gilman, 2002 (joint position sense); Campbell & Barohn, 8e (2020).

Clinical Tests and Assessment Methods

Separate the cognitive/mental-status questions from sensory and coordination tests.

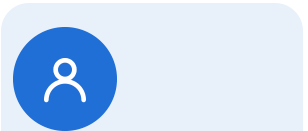
A ORIENTATION ASSESSMENT

Ask

-  Name
-  Location
-  Date / day / month / year
-  Current situation / reason for being there

Observe

-  Attention
-  Understanding of questions
-  Appropriate answers
-  Consistency of responses

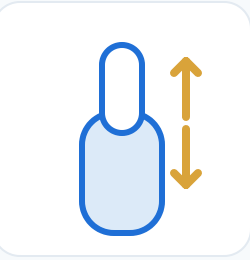
**Person**

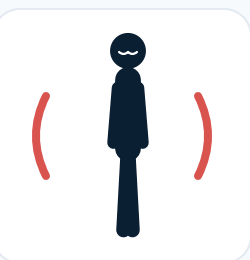
**Place**

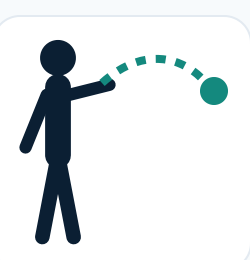
**Time**

**Situation**

B BODY POSITION / PROPRICEPTION / COORDINATION

**JOINT POSITION SENSE**
Eyes closed → move finger/toe up or down → “Up or down?”

**ROMBERG**
Feet together, eyes open → close eyes → watch for increased sway / loss of balance.

**FINGER-NOSE**
Finger to own nose → examiner’s finger → back again. Assesses coordination.

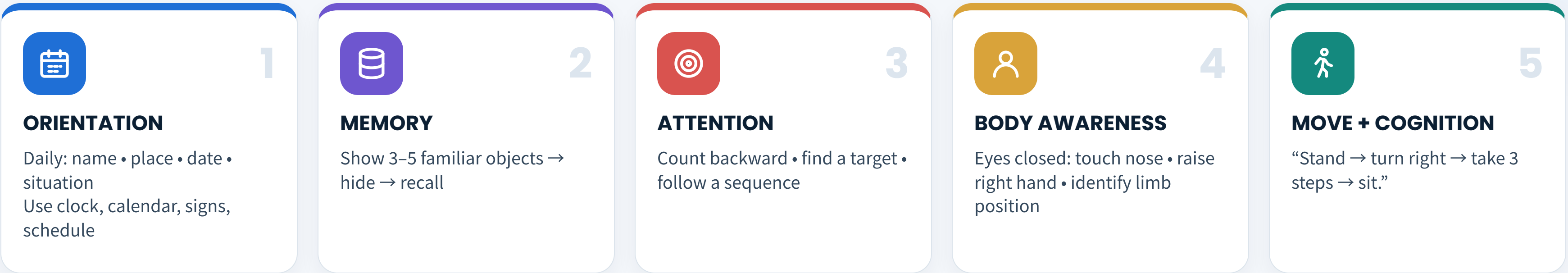
Orientation = cognitive / mental status Joint position sense = proprioception Finger-nose = coordination

 **Safety:** perform balance tests with appropriate support / supervision to reduce fall risk.

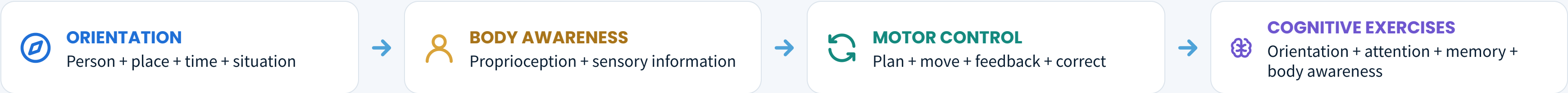
Sources: Campbell & Barohn, 8e (2020); NCBI Bookshelf, Romberg Test (updated 2026); Neurologic Exam (2026).

Exercises to Support Orientation, Body Awareness and Cognitive Function

Use rehabilitation / stimulation activities that fit the patient’s diagnosis, goals and safety needs.



TAKE-HOME FLOW



Final message: The brain integrates mind, body and environment to maintain awareness and guide appropriate behaviour and movement.

Sources: ASHA Clinical Practice Guideline, AJSLP (2022); INCOG 2.0 cognitive rehabilitation guidance (2023); Jak, Neurologic Clinics (2024).



Thank You

Questions & discussion

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